DATA SCIENCE & ANALYTICS · SESSION 06

# Sales Performance Analysis

## Use Case 1: putting the whole pipeline to work on a real retail dataset

Asst.Prof.Dr. Chaiwat N. · 2026

2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

# Session Agenda

01 **The Business Question**
The retail brief and the dataset

02 **Load & Audit**
Inspect, quality-check, add time features

03 **KPIs**
Revenue, growth, margin, top and bottom

04 **Time-Series Trends**
Monthly aggregation, rolling averages, seasonality

05 **Segments**
Cutting the numbers by category and region

06 **Workshop Block**
Add and defend your own KPI

01

PART 01

# The Business Question

A mid-sized retailer has two years of transactions. Leadership wants three answers: where is the business growing, where is it losing, and what next?

# Use Case 1 is the first time the whole pipeline runs on a real business problem

## The brief

| | |
|---|---|
| **The client** | A mid-sized retailer, two years of transaction data: date, category, product, price, margin, and quantity. |
| **The questions** | Where is revenue growing? Where is it shrinking? Which products and categories drive the result — and what should we do? |
| **Your job** | Load, audit, compute KPIs, find trends, and end with a recommendation a manager could act on Monday morning. |
| **Why it matters** | Sessions 1–5 taught the moves; this is the first session where they combine into a real deliverable. |



Source: DS Course, Session 6, 2026

02

PART 02

# Load & Audit

Before any KPI, confirm the data is sound: shape, types, missing values, duplicates, value ranges — then enrich it with time features.

# Load and audit: never compute a KPI on data you haven't checked

## Trust, then calculate

### Inspect

shape, head, info, describe — the Session 1 ritual. Confirm 5,000 rows and the columns you expected.

### Quality-check

Missing values, duplicate transactions, and impossible values (negative price, zero quantity) before anything else.

### Add time features

Derive year, month, and quarter from the date so you can aggregate and find seasonality later.

**PYTHON · LOAD & AUDIT**

```
df = pd.read_csv('retail.csv',
                 parse_dates=['date'])
df['revenue'] = df['price'] * df['quantity']

# quick audit
print(df.shape)
print(df.isnull().sum())

# time features for aggregation
df['year']  = df['date'].dt.year
df['month'] = df['date'].dt.month
```

revenue = price x quantity — the column you'll use all session.

# KPIs: the handful of numbers leadership actually asks for

## From rows to metrics

### Revenue by period

groupby year or month and sum revenue. The headline number every other KPI is measured against.

### Growth rate

pct_change() on the period totals turns levels into the growth story — up or down, and by how much.

### Margin & ranking

Revenue × margin gives profit; sort to surface the top and bottom products and categories.

**PYTHON · KPIs**

```
# revenue by year
rev = df.groupby('year')['revenue'].sum()

# year-over-year growth
rev.pct_change().round(3) * 100

# top 10 products by revenue
(df.groupby('product')['revenue'].sum()
   .nlargest(10))
```

pct_change() turns levels into the growth narrative.

03

PART 03

# KPIs

Revenue, growth rate, profit margin, and product rankings — the four numbers that answer 'where are we growing and losing?'

# Four KPIs answer most of the brief — compute them, then read them together

## The metric set

| | |
|---|---|
| **Revenue** | Total and by period. The scoreboard. Always the first KPI, because everything else is relative to it. |
| **Growth rate** | Period-over-period percentage change. Direction and momentum — a falling growth rate warns before revenue does. |
| **Profit margin** | Revenue weighted by margin. High revenue at thin margin can lose to lower revenue at a healthy one. |
| **Top & bottom** | The products and categories driving and dragging the result. Where the recommendation will ultimately point. |



Source: DS Course, Session 6, 2026

04

PART 04

# Time-Series Trends

Aggregate to a monthly rhythm, smooth the noise with a rolling average, and expose the seasonality hiding in the daily data.

# Trends over time: aggregate, smooth, and expose seasonality

## See the rhythm

### Monthly aggregation

Resample or group to month-level revenue. Daily data is too noisy to see the trend; monthly reveals it.

### Rolling average

A 3-month rolling mean smooths spikes so the underlying direction is visible. The classic trend line.

### Seasonality

A month × year pivot, drawn as a heatmap, shows which months repeat strong or weak — the seasonal pattern.

**PYTHON · TRENDS**

```
monthly = (df.groupby(['year','month'])
             ['revenue'].sum())

# 3-month rolling average
monthly.rolling(3).mean()

# seasonality grid: month x year
df.pivot_table(index='month',
    columns='year', values='revenue',
    aggfunc='sum')
```

Aggregate to month, then smooth to see the trend.

05

PART 05

# Segments

The headline number hides the story. Cut revenue and growth by category and region to find where the business is really winning and losing.

# Averages hide the action — segment to find where growth and loss actually live

## Cut the numbers

### By category

Total revenue can be flat while Electronics booms and Clothing sinks. The segments tell opposite stories.

### By region

The same product can grow in one region and shrink in another. Geography often drives the recommendation.

### Cross-cut

Category × region together pinpoints the exact cell that's moving — far sharper than either alone.

### Then act

A finding at segment level — 'Electronics in the North is the growth engine' — is something leadership can act on.



Source: DS Course, Session 6, 2026

06

PART 06

# Workshop Block

Add one KPI of your own to the analysis, compute it correctly, and write the one-sentence insight it produces.

# By the end of this block, every student has added and defended one original KPI

**Student Exploration · 20 Minutes**

| | |
|---|---|
| **Choose a KPI** | Something the brief implies but we didn't compute — average order value, repeat-purchase rate, revenue per region. |
| **Compute it** | Use the groupby and aggregation moves from today. Confirm the number is sensible against the totals. |
| **Visualize** | One clean, titled chart that shows what your KPI reveals across time or segment. |
| **One-sentence insight** | State what it means for the business. The KPI plus its insight sentence is your deliverable. |



Source: Workshop, Session 6, 2026

KEY TAKEAWAY

# A KPI is only useful when it ends in a decision.